

# GAIRA V7 — Phase 07

## BSV2 — the biochemical programme layer

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What biochemical programmes best explain the validated chemistry evidence?

|                                    |                                                                   |
|------------------------------------|-------------------------------------------------------------------|
| Status                             | COMPLETE — BSV2 adopted: True                                     |
| Input                              | Chemistry Evidence 375 x 16, and nothing else                     |
| Selected                           | semi_nmf at K = 9 (pre-registered rule, unadjusted)               |
| Compression                        | 1.8x (16 → 9)                                                     |
| Reconstruction                     | EV 0.818 · cosine 0.929 · RMSE 0.080                              |
| Held-out reconstruction            | EV 0.765 (gap +0.053)                                             |
| Stability                          | bootstrap 0.972 · seed 0.957 · fold 0.979                         |
| Programmes below 0.70 recovery     | 0 of 9                                                            |
| Information retained               | 0.818 (floor 0.50)                                                |
| Noise robustness                   | programme cosine 0.948                                            |
| Genuine multi-chemistry programmes | 7 of 9 — the rest are near-single-class                           |
| Programme redundancy               | max pairwise overlap 0.400                                        |
| P-02-compliant alternative         | orthogonal_nmf K=6, objective cost +0.035                         |
| Frozen inputs                      | LSM 208482d6... · CSM 0b4aa550... · engine 20d8bd99... — verified |
| Phase 08                           | NOT BEGUN                                                         |
| Completed                          | 2026-08-07 03:54:12 UTC                                           |

# Figures

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01 architecture

02 k sweep

03 programme loadings

04 overlap

05 activation map

06 representative molecules

07 programme spectra

08 stability

09 noise robustness

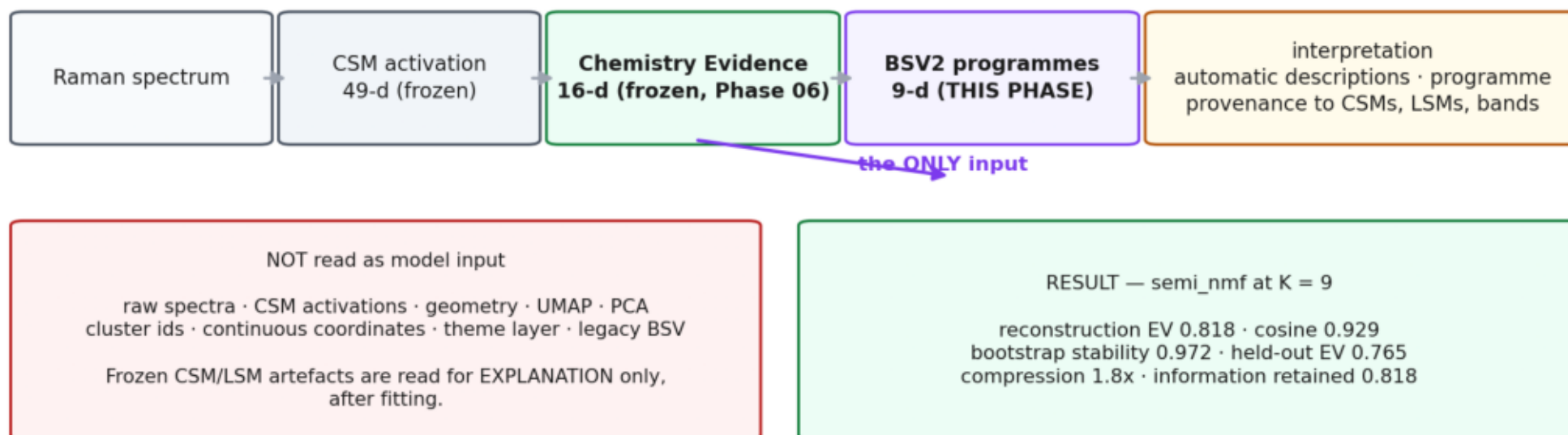
10 reconstruction

11 programme graph

12 comparison

Figure 01

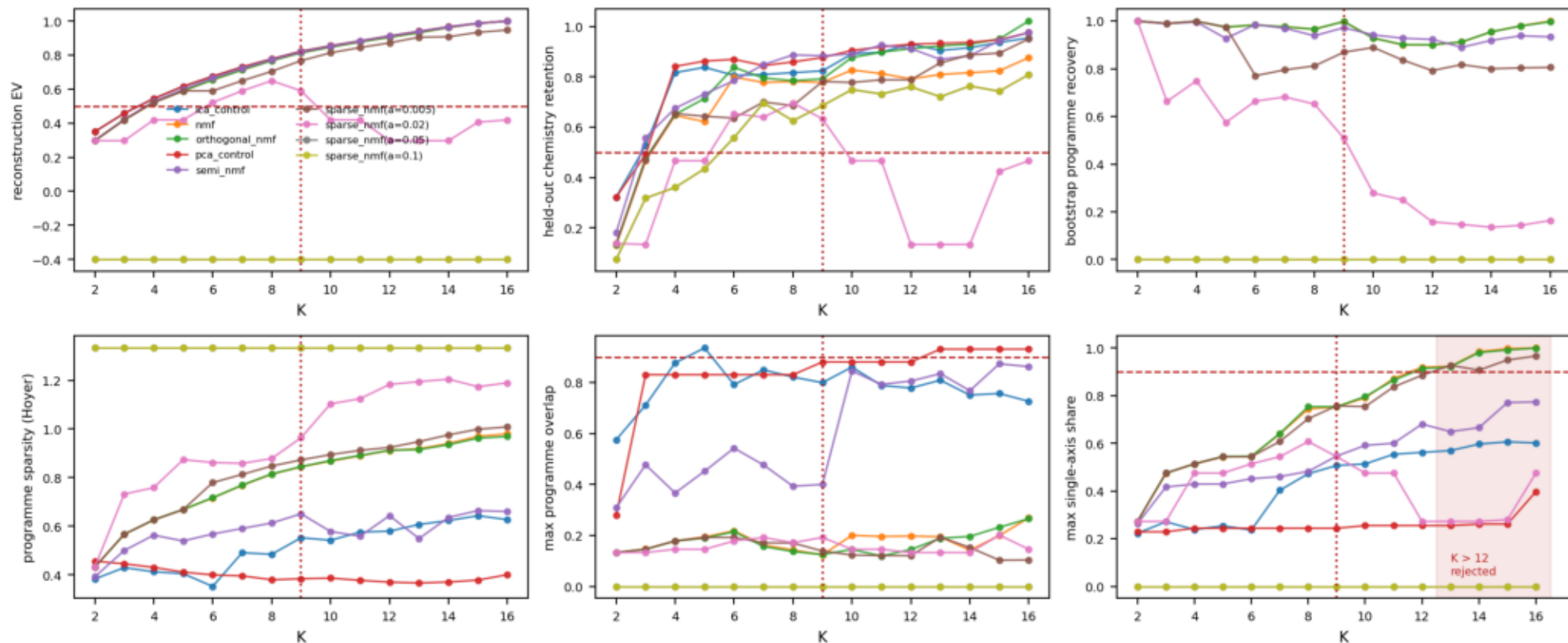
Figure 1 · Phase 07 architecture — Chemistry Evidence → BSV2 → interpretation



Phase 07 architecture. BSV2 is learned from the validated 16-dimensional Chemistry Evidence matrix and nothing else: no spectra, no CSM activations, no geometry, no cluster ids, no coordinates, no theme layer, no legacy BSV. Frozen CSM and LSM artefacts are read for explanation only, after fitting.

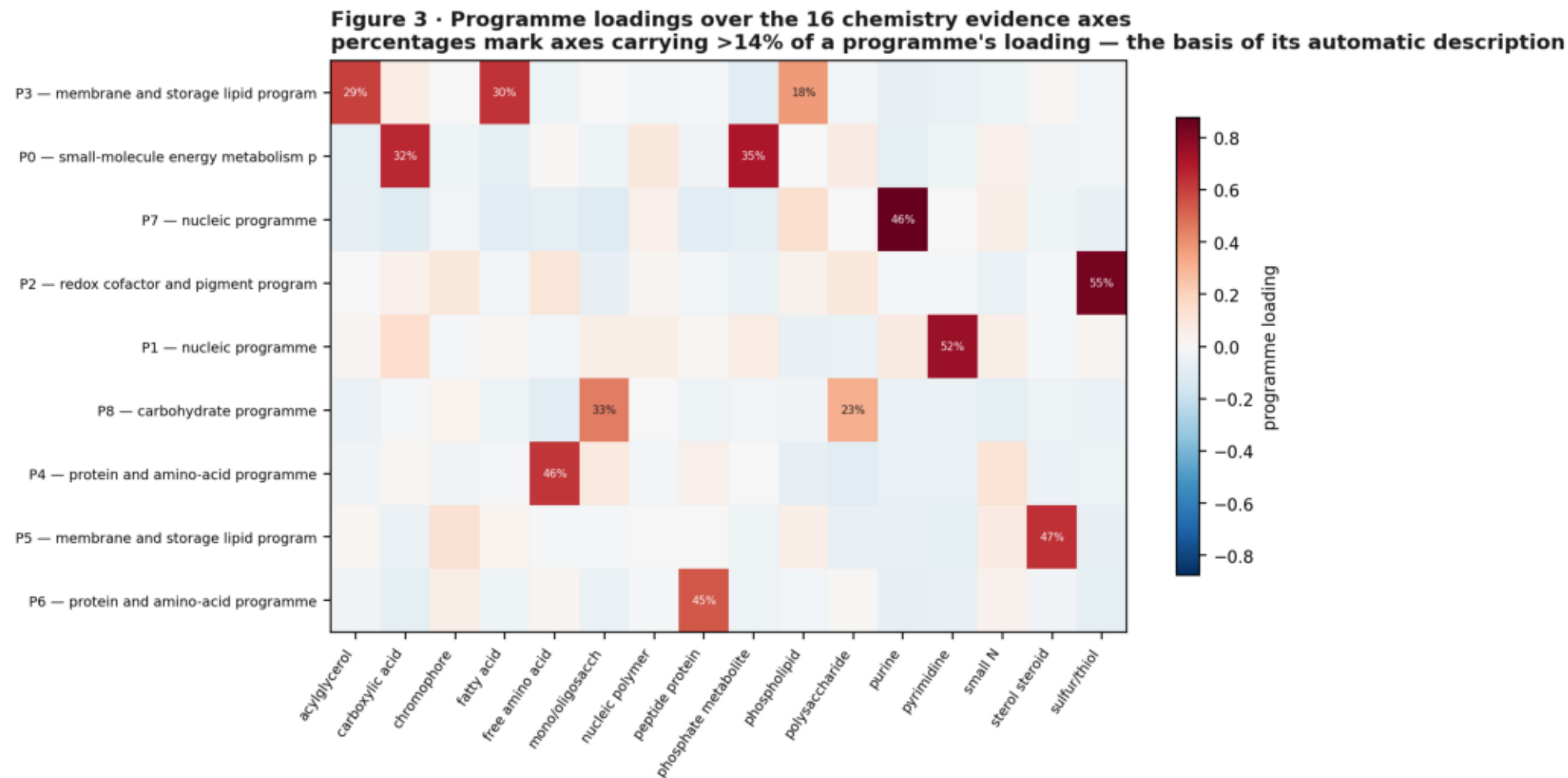
**Figure 02**

**Figure 2 · Programme-number sweep — 6 families × K = 2...16, with the pre-registered floors decision: semi\_nmf at K = 9 · 38 of 135 candidates eligible · red dotted line = the selected K**



Programme-number sweep: six model families over K = 2 to 16, with the pre-registered floors drawn. The floors on K ( $\leq$  the input's effective rank) and on single-axis share were added after a first run selected K = 16, where NMF learns a permutation of the identity and every programme equals one chemistry class.

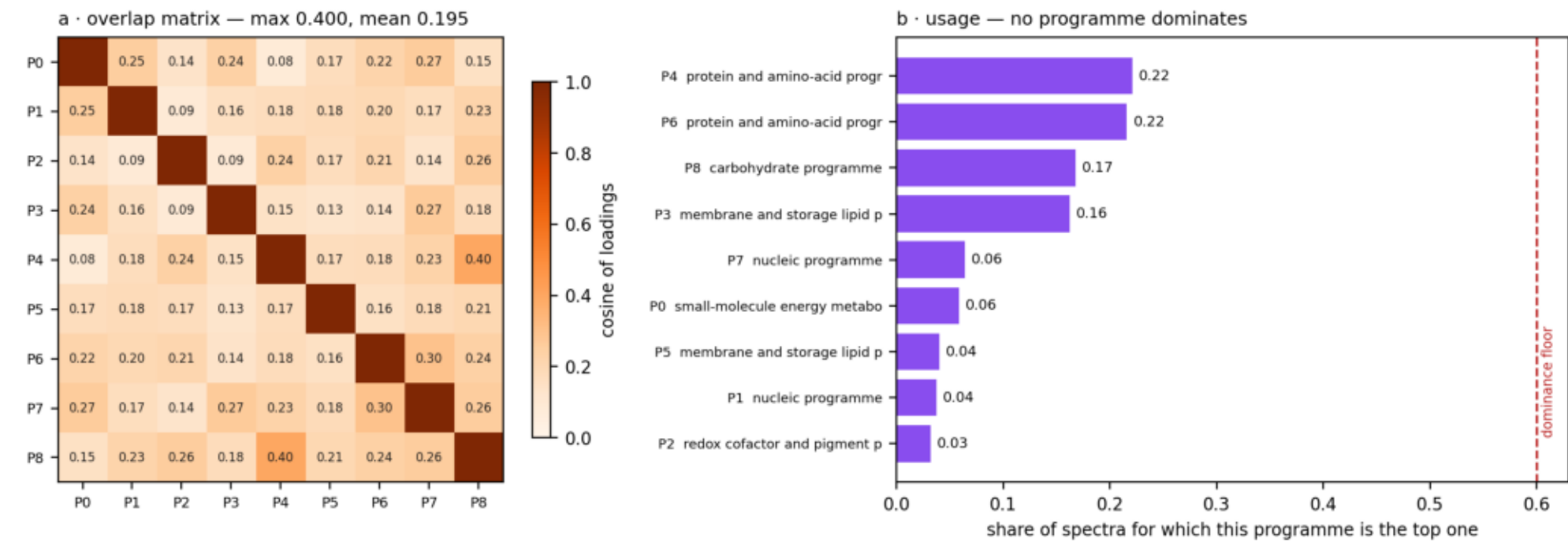
Figure 03



Programme loadings over the 16 chemistry axes, with the percentages that generate each programme's automatic description. Three programmes are genuine multi-chemistry compressions - lipid, energy metabolism, carbohydrate - and six are near-single-class.

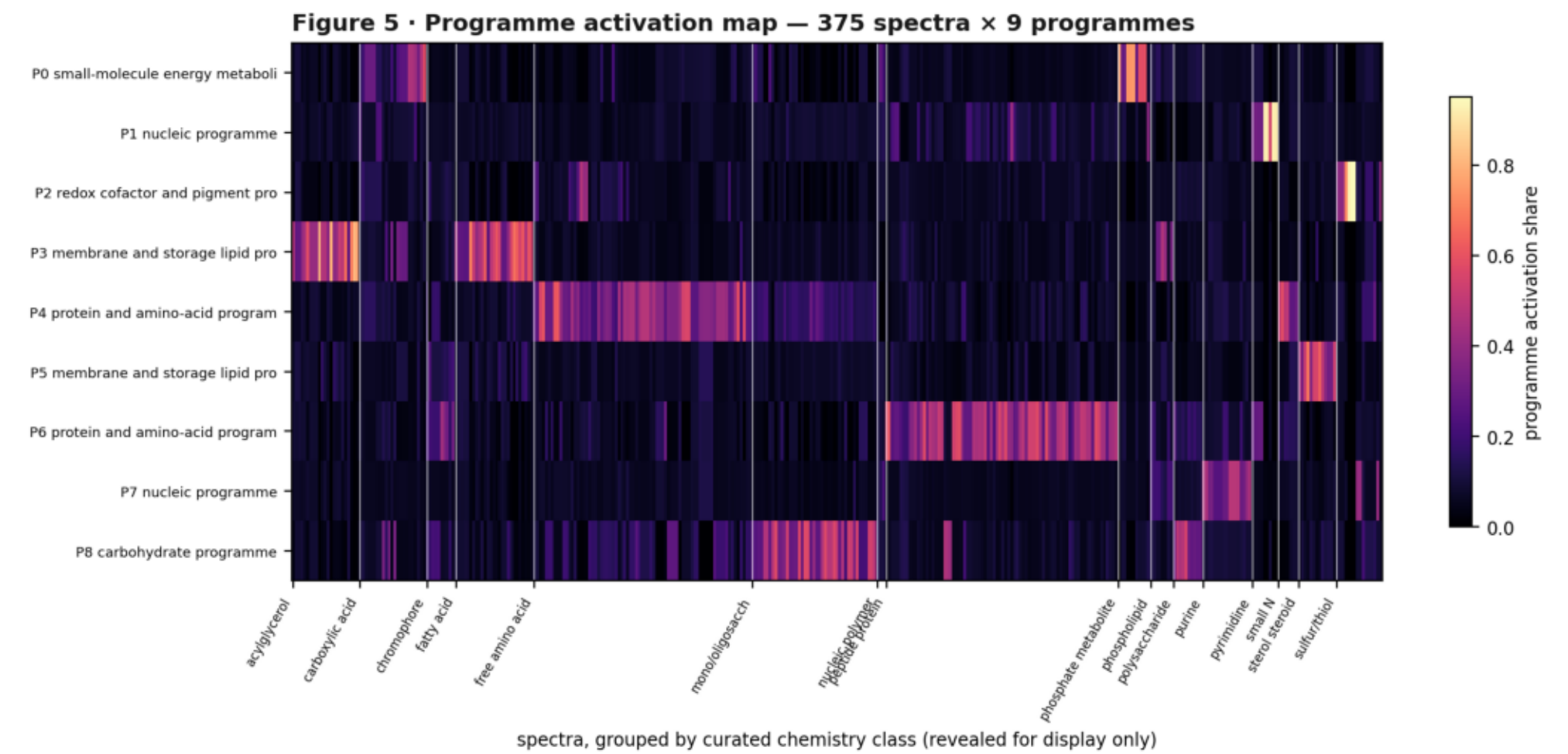
Figure 04

Figure 4 · Programme overlap and usage



Programme overlap and usage. No two programmes duplicate each other and none dominates: the top programme wins for 22% of spectra against a 60% ceiling.

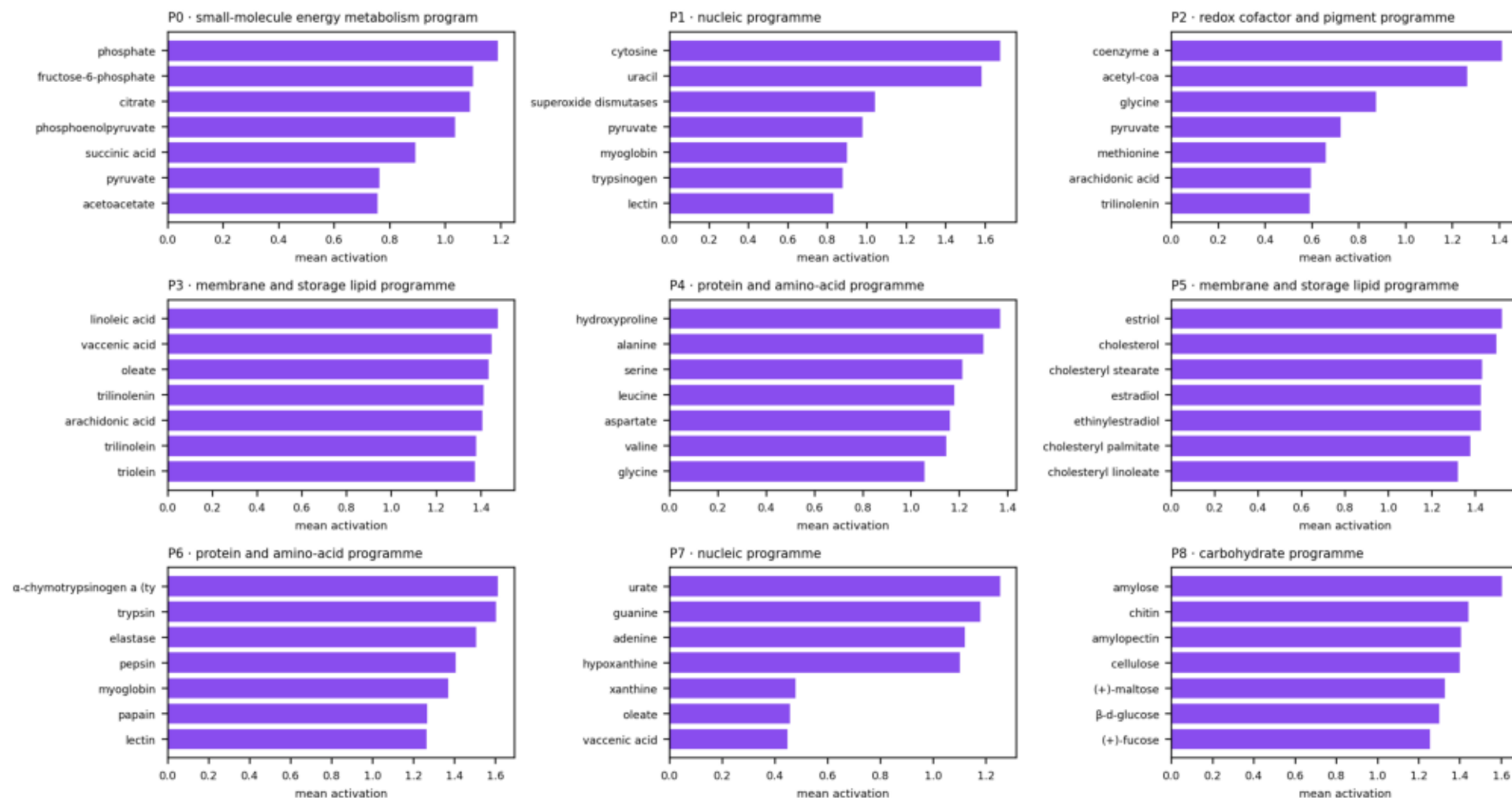
Figure 05



Programme activation map: 375 spectra by 9 programmes, spectra grouped by curated chemistry class. The class labels are revealed for display only and were never an input to the factorisation.

**Figure 06**

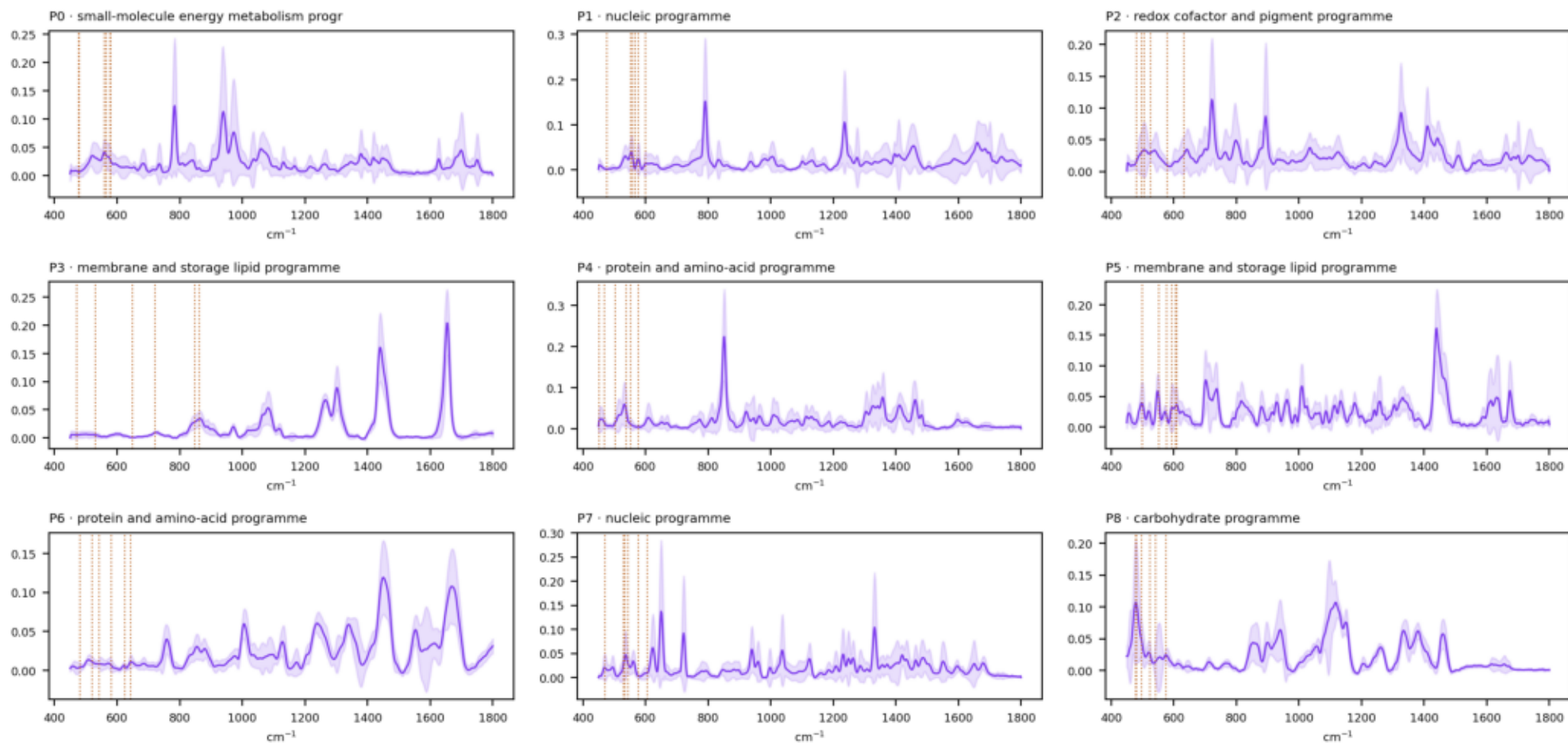
**Figure 6 • Representative molecules per programme — the molecules that activate it most**



The molecules that activate each programme most. Evidence first: these lists are what the automatic descriptions are composed from.

**Figure 07**

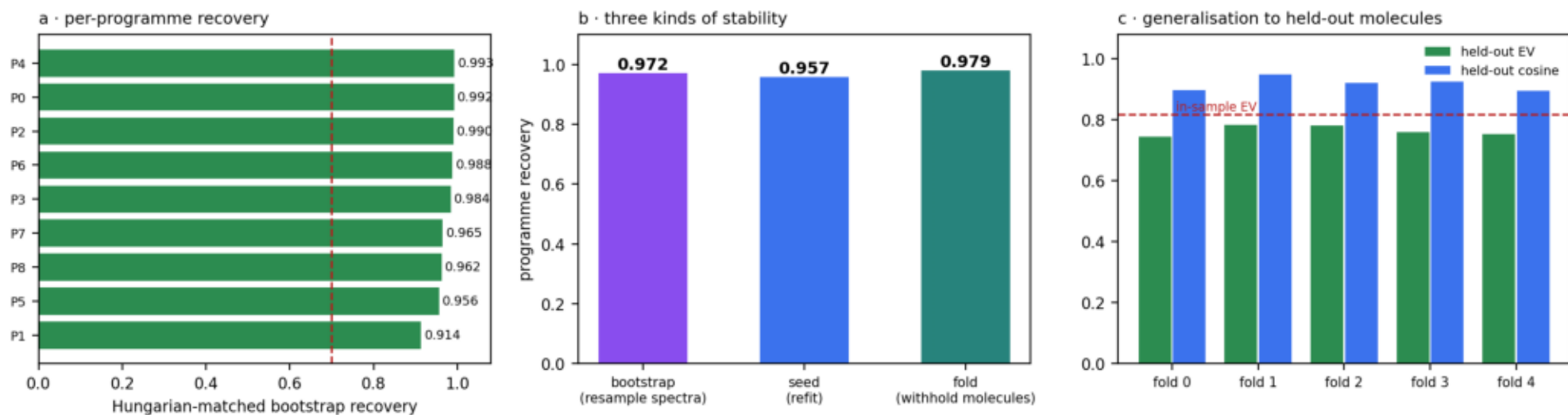
**Figure 7 · Mean Raman spectrum  $\pm 1$  sd of each programme's top molecules (amber = the programme's dominant CSM bands) spectra are shown for interpretation only — they were never an input to the factorisation**



Mean Raman spectrum of each programme's top molecules, with the programme's dominant CSM bands marked. Spectra are shown for interpretation only.

**Figure 08**

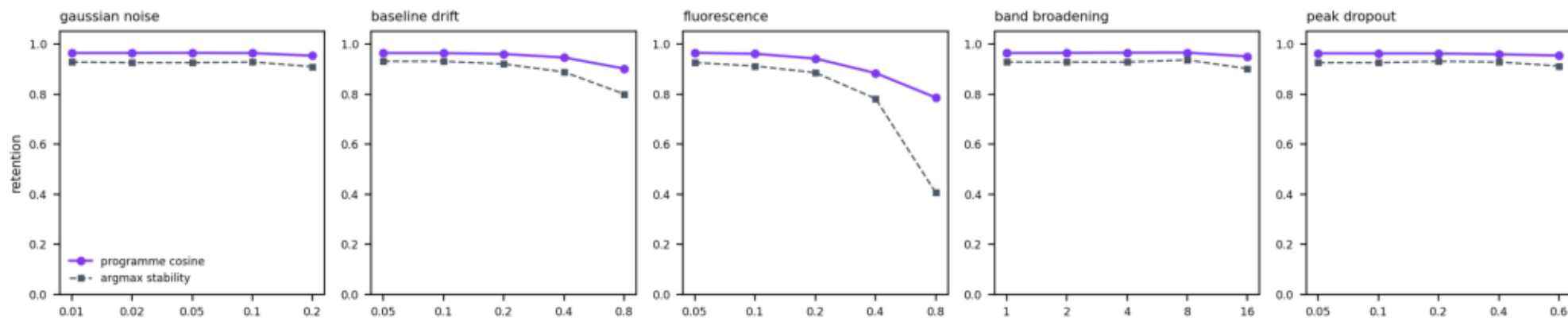
**Figure 8 · Programme stability and generalisation**



Stability under three separate perturbations - resampling spectra, refitting from a new seed, and withholding a whole molecule-grouped fold - and generalisation to held-out molecules.

**Figure 09**

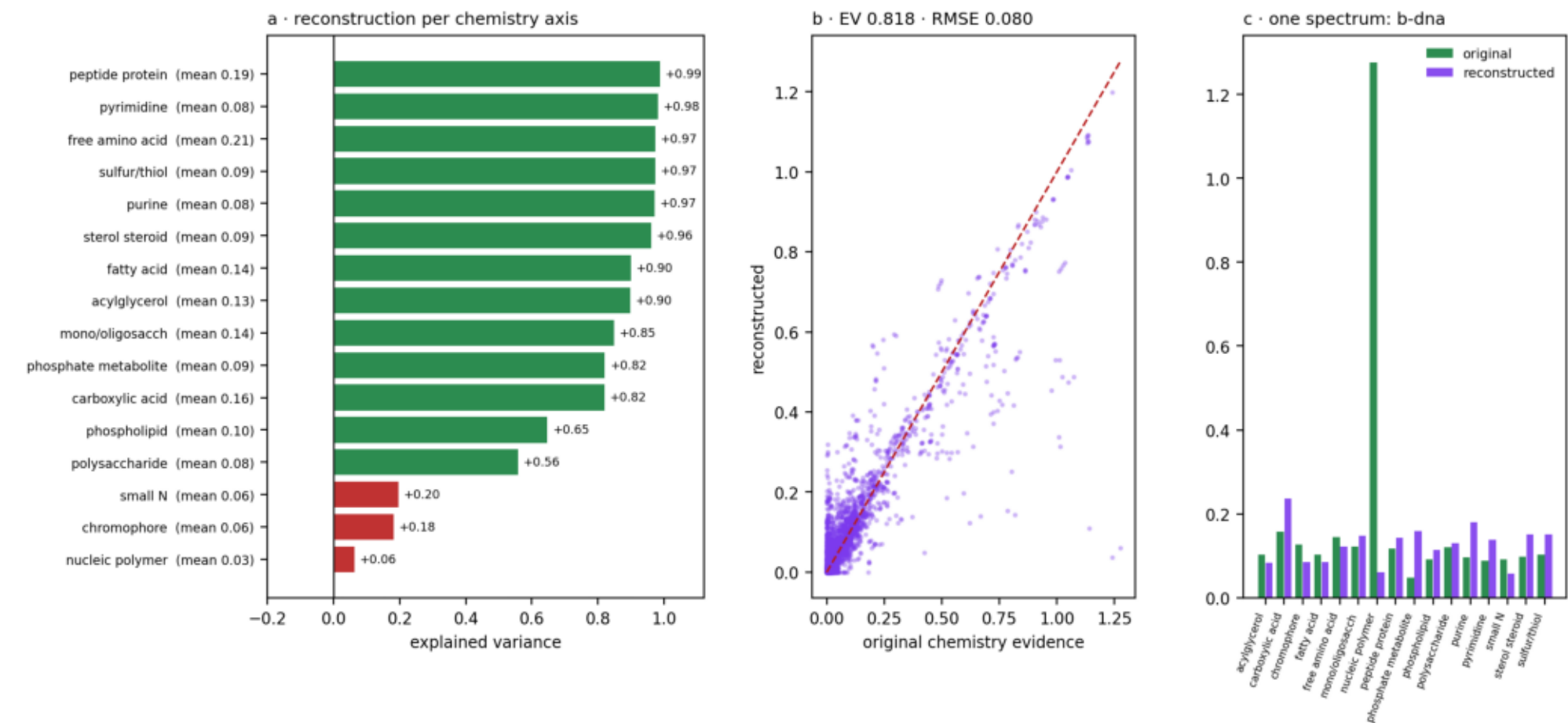
**Figure 9 · Noise robustness propagated through the whole frozen chain (spectrum → CSM → Chemistry Evidence → BSV2)  
mean programme cosine 0.948, argmax stability 0.889**



Noise robustness propagated through the entire frozen chain: spectrum to CSM to Chemistry Evidence to BSV2. This is the only place in the phase where spectra are touched, and only to perturb them.

Figure 10

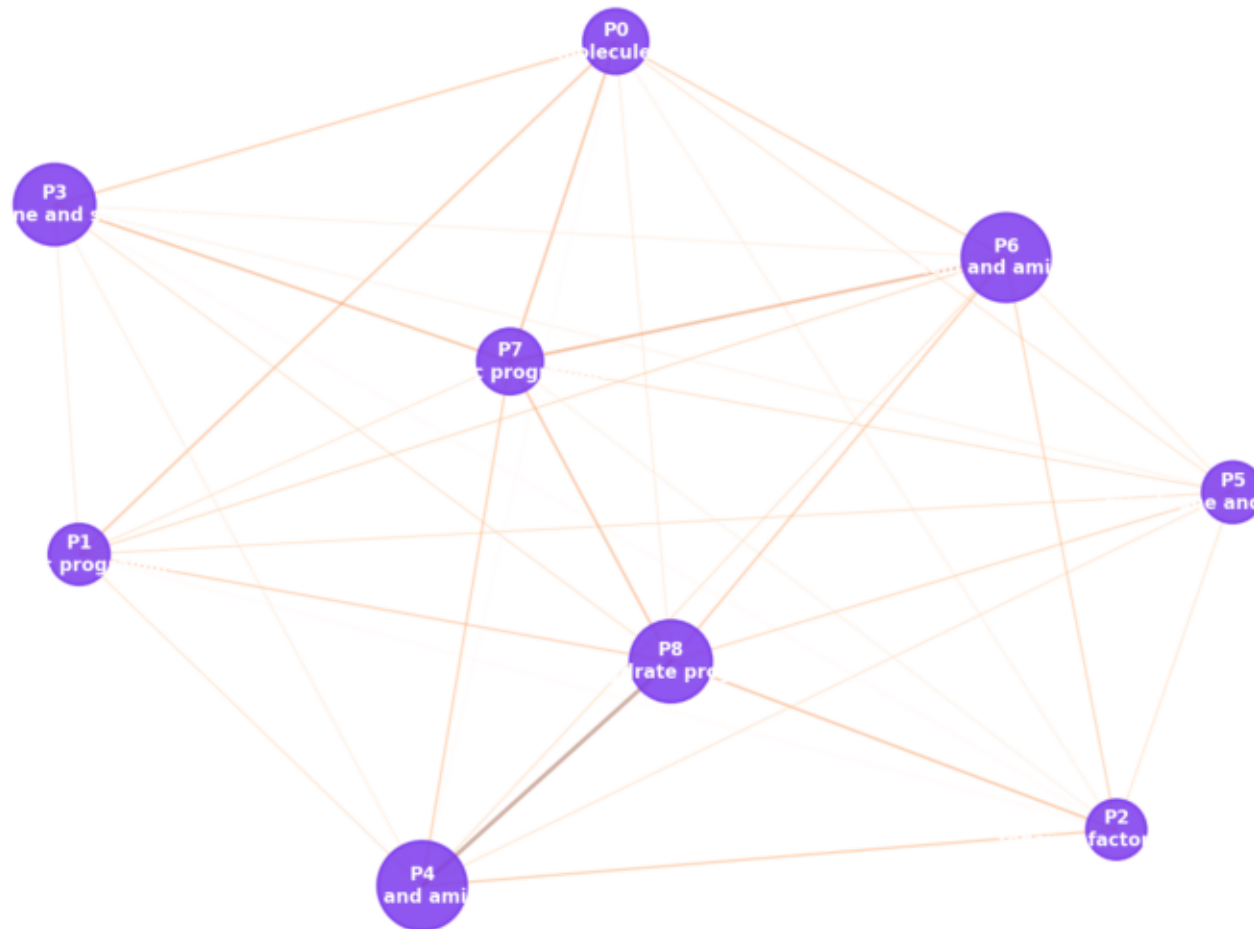
Figure 10 · Can BSV2 reconstruct the Chemistry Evidence vector?



Reconstruction of the Chemistry Evidence vector, globally and per chemistry axis. The axes reconstructed worst are the ones with the least evidence to begin with.

**Figure 11**

**Figure 11 · Programme similarity graph — edge width is loading cosine, node size is usage share  
edges shown above 0.05; max overlap 0.400**

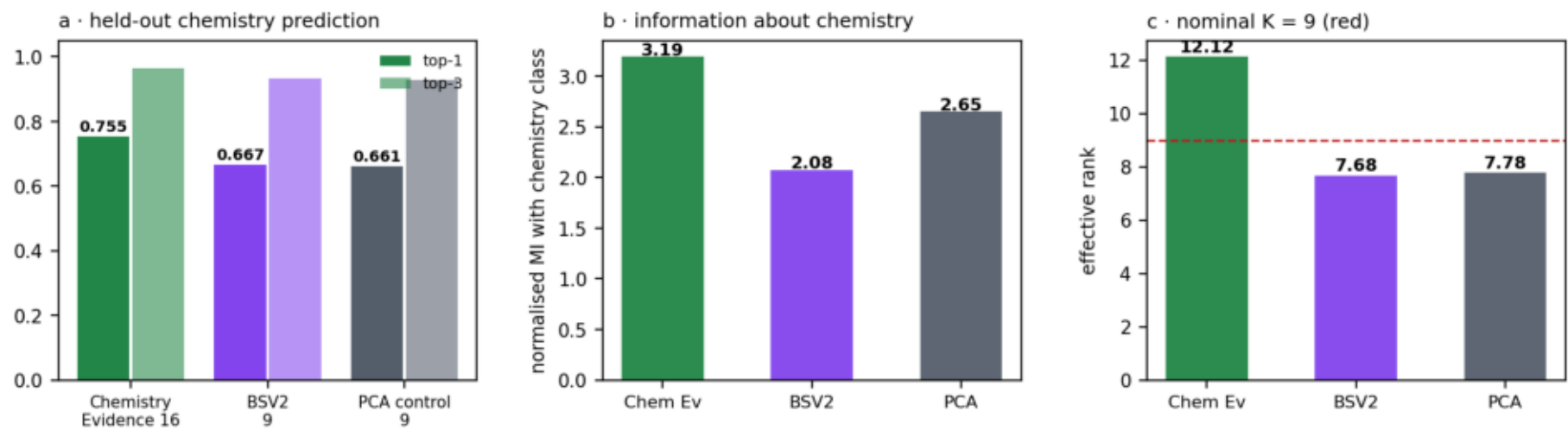


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Programme similarity graph. Edge width is the cosine between programme loadings, node size is usage share.

Figure 12

Figure 12 · Chemistry Evidence versus BSV2 versus a PCA control, and the decision gate



|                                                         |                                  |                                                   |                                                                |                                                                       |                                                        |
|---------------------------------------------------------|----------------------------------|---------------------------------------------------|----------------------------------------------------------------|-----------------------------------------------------------------------|--------------------------------------------------------|
| G1 frozen fingerprints verified                         | G2 no upstream artifact refitted | G3 input is Chemistry Evidence only               | G4 no geometry, cluster ids or coordinates used                | G5 Raman-only scope                                                   | G6 all 6 families x 15 K fitted, none silently dropped |
| G7 selection rule pre-registered and applied unadjusted | G8 K not chosen by hand          | G9 programme naming automatic, none named by hand | G10 soft membership — no programme is a single chemistry class | G11 activations non-negative                                          | G12 generalisation measured on held-out molecules      |
| G13 noise robustness measured through the frozen chain  | G14 deterministic on rerun       | G15 PNG-only figure policy declared               | G16 Phase 08 not begun                                         | G17 the P-02 loading-sign question is reported, not silently resolved |                                                        |

17 of 17 gates pass · adopted semi\_nmf K=9 · compression 1.8x · P-02-compliant alternative orthogonal\_nmf K=6 (objective +0.035)

Chemistry Evidence versus BSV2 versus a PCA control at the same K, and the decision gate.